

Ashley Clayton

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<https://github.com/claytonclayton>

EDUCATION

University of New South Wales

Bachelor of Computer Science

Kensington, Sydney

Feb 2022 – Dec 2024

University of Technology Sydney

Bachelor of Computer Science - Undergraduate IT Dean's List

Chippendale, Sydney

Feb 2021 - Dec 2021

WORK EXPERIENCE

Lab Demonstrator

Feb 2024 - Present

COMP3121 - Algorithm Design and Analysis, COMP4128 - Programming Challenges

UNSW

- Attained a student approval rating 10% higher than that of my peers, by tailoring explanations to student ability and clarifying conventions for academic algorithmic proof writing.
- Contributed original problems to 66% of COMP4128 (2024) assessable exams by showcasing three competitive programming problems spanning distinct algorithmic paradigms and validating solution approaches with tutors.

Problem Setter, Judge

May 2025 - Sep 2025

Prog Comp

UNSW

- Proposed, developed and adjudicated problems as a contracted problem setter for a statewide, high school competitive programming competition involving more than 120 teams.
- Diagnosed and resolved a live contest issue by reverting affected Git commits and issuing clarification notes to teams onsite.

VOLUNTEERING

Director of Competitions

Nov 2023 - Dec 2025

CPMSoc - Competitive Programming and Mathematics Society

UNSW

- Increased society funding by \$2000 per trimester by organising sponsored competitions in collaboration with IMC, Citadel Securities, and Jane Street, facilitating recruiter-student engagement.
- Assumed primary responsibility for contest delivery while the co-director was on exchange, more than doubling personal problem-writing output and overseeing testing, scheduling and prize acquisition for the 2024 Term 3 Programming and Mathematics O-Week Contest.

PROJECTS

Resume Parser | *Haskell, Megaparsec, LaTeX, React*

Oct 2025 - Present

- Developing a web application and Cli that parses and transpiles a Markdown sub-language into Latex to create aesthetically pleasing resume PDFs.
- Utilises ParsecT, ReaderT and ExceptT monad transformer stacks to constrain effects and explicitly model syntax and runtime errors, improving correctness and testability.

Clayton's Problems | *C++, Bash, Python, Haskell*

Feb 2024 - Dec 2025

- <https://github.com/claytonclayton/claytons-problems>
- Curated a collection of 25 original competitive programming problems, including statements, solutions, explanations, and randomised test case generation and validation.
- Built a Bash-based evaluation tool to visualise accepted, incorrect, and time limit exceeded submissions across C, C++, Python, and Haskell solutions.

TECHNICAL SKILLS

Languages: Haskell, C/C++, Python, Bash, Java, Typescript, LaTeX, PostgreSQL, MIPS Assembly

Libraries: testlib, Megaparsec, matplotlib, React

Developer Tools: Git, Docker, neovim, Ubuntu